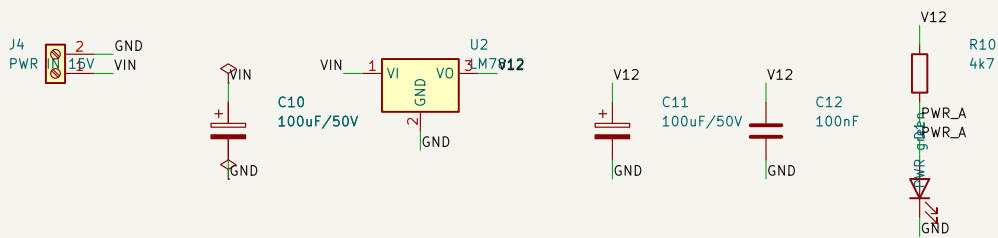


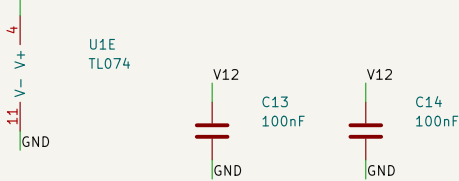
Bose Companion 5 sub crossover -- mono--summing 2nd-order Sallen-Key low-pass.

Single +12 V rail, no negative supply. All AC referenced to VGND = 6 V (A3 buffers the R8/R9 mid-rail divider).

A POWER IN --> +12 V



B U1_SUPPLY

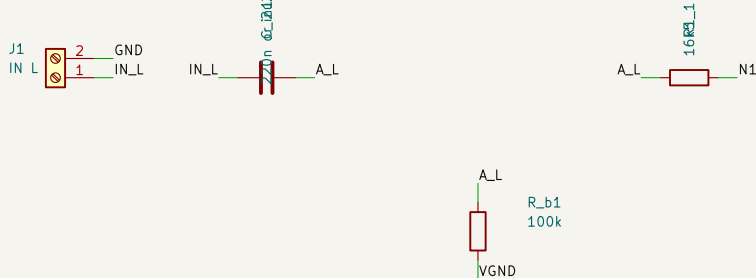


C VIRTUAL GROUND 6 V



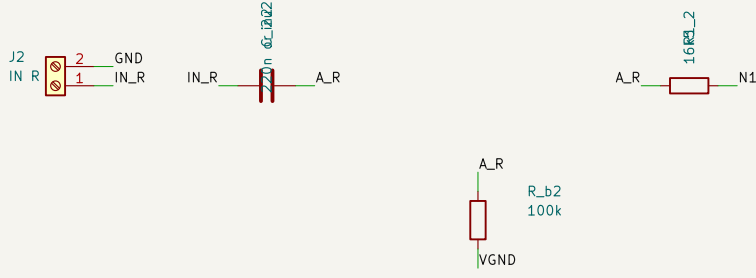
U1 supply: pin 4 = V12, pin 11 = GND. No negative rail.

D INPUT L

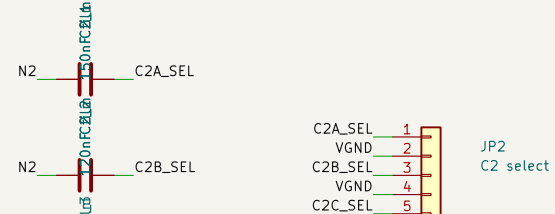
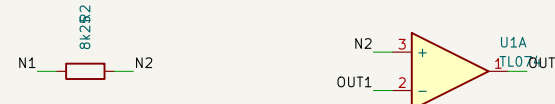
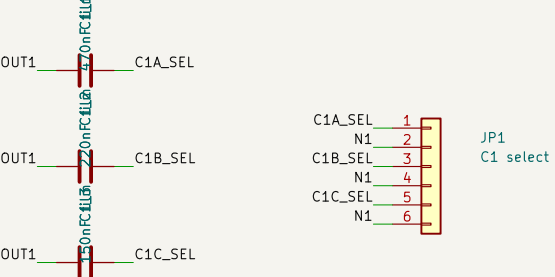


R_b1/R_b2 are the only DC path to A1's + input (C_in blocks DC, C2 is a capacitor). Without them A1 drifts to a rail. They sit after the coupling caps -- 100k at N1 would shift every corner by 8%.

E INPUT R



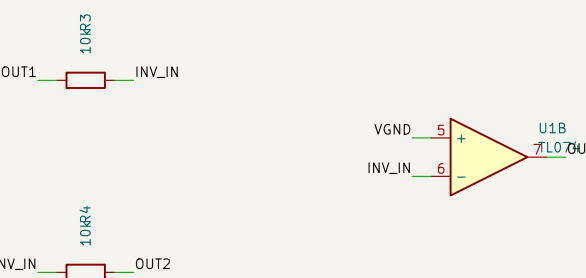
F SALLEN-KEY LOW-PASS



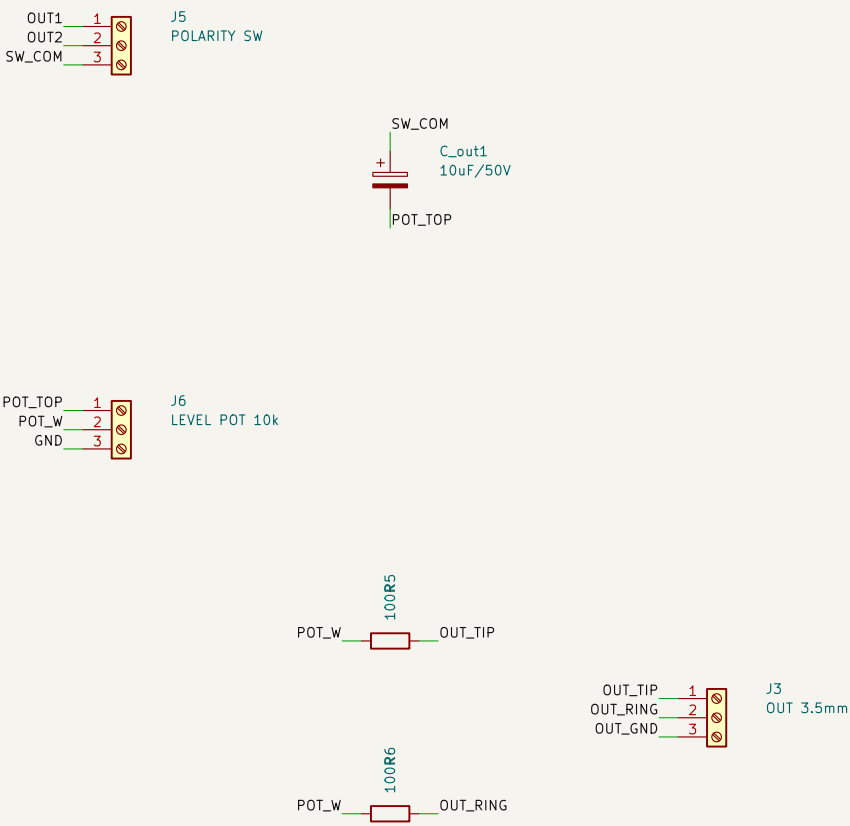
C1 returns to N1 -- the Sallen-Key feedback path. Not N2. JP1 selects C1 (odd pin = cap, even pin = N1). JP2 selects C2 (odd = cap, even = VGND). C2s select INDEPENDENTLY -- nine corners, not three pairs. Corners below are MEASURED on the rev A board, not calculated: the quoted $f_0 = 1/(2\pi R \sqrt{C1 \cdot C2})$ is not a -3 dB point, and C_in sits inside the filter. See Results - Sub Crossover Bring-up.

Intended use is a panel 2-pole 3-position rotary, one pole per header:
pos 1 94 Hz A0->JP1 even, A1->JP1.1 (470n) B0->JP2 even, B1->JP2.1
pos 2 136 Hz A2->JP1.5 (150n) B2->JP2.3
pos 3 189 Hz A3->JP1.3 (220n) B3->JP2.5
NEVER wire one header pin to two switch lugs unless both lugs want the same capacitance -- the lugs are then tied together and every position collapses to the same value. This is why C1.1 is fitted rather than paralleling C1.2 || C1.3 on one lug.
Shunts across a pin pair still work as a bench fallback.

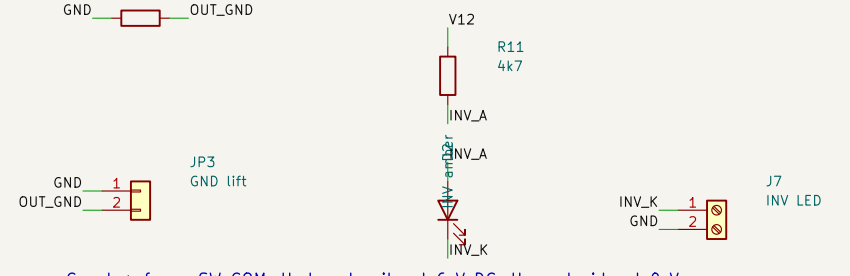
G POLARITY INVERTER



H OUTPUT STAGE



I STATUS LEDs



C_in1/C_in2 use a dual-pitch footprint: 5.00 mm for the 220n as built (or a 2u2 non-polarised electrolytic), 15.00 mm for a real 2u2 film part. Fit ONE pair. 220n puts -8.8k of reactance in series with R1 at 82 Hz, so it is part of the filter, not just an Input high-pass -- it costs 1.3-3.1 dB of passband level and damps the peaking. Whether that is a defect or a feature is a listening question; Gate 11 has not answered it, so both pitches are on the board and the envelope reserves room for the larger part either way.

Off-board: level pot (J6), polarity switch (J5), all panel jacks (J1/J2/J3).
Process: single-sided THT, bottom copper only, board outline 100 x 100 mm. Copper is isolation-milled on the CNC with an 0.8 mm flat end mill; the fiber laser does the silkscreen only. Routing clearance 0.85 mm / track 1.0 mm. The 0.8 mm tool width is a hard floor on every copper-to-copper gap. Fixed-pitch pads sit below the 0.85 mm target and carry a kicad_dru exception: DIP-14 0.94 mm (passes), 2.54 mm headers and the TD-220 0.84 mm (do not). Drills: 0.8 mm x52, 1.0 x26, 1.1 x3, 1.2 x4, 1.5 x15.

Reference names vs the handoff document: KiCad treats a reference with no trailing digit as unannotated, so C1a/C1b/C1c -> C1.1/C1.2/C1.3, C2a/C2b/C2c -> C2.1/C2.2/C2.3, R1a/R1b -> R1.1/R1.2, C_out -> C_out1.

Single-sided THT, CNC isolation milling (0.8 mm end mill), laser silkscreen
Mono--summing 2nd-order Sallen-Key low-pass, single +12 V rail